

Trading Strategies Reference Catalogue

Every signal-generating strategy implemented in the project — NIFTY index-options engines, the paper-trading challenges, the book-based comparison suite, and the equity / SIP screeners.

Generated from source: [internal/naren/analysis/](#) (options & alpha packages)

Scope: index-options strategies, scalping system, daily classifier, 7 book strategies, Triple-Axiom config, and 5 equity screeners.

Contents

1. NIFTY Options Challenge Engine — the 90-day OPTIONS regime model
2. Scalping Challenge — the 90-day SCALP 1-minute system
3. Daily Classifier — EMA/ATR regime model for long backtests
4. Book-Based Strategy Suite — 7 published-book strategies for comparison
5. Triple-Axiom — SuperTrend + EMA pullback configuration
6. Equity & SIP Screeners — intraday, BTST, investment, undervalued, US SIP

1 · NIFTY Options Challenge Engine

The core engine behind the 90-day OPTIONS paper/live challenge (**options.Analyze**). It reads 5-minute NIFTY index candles, scores a market regime, then picks one of seven strategy structures via a priority decision tree. Trend is judged on EMA9 / EMA21 / EMA50 stacking, with RSI-14 momentum, a VWAP proxy, ATR-14 volatility and an ATR-scaled trend-strength score (0–100). For the challenge itself only directional CE/PE buys are taken; the neutral structures are signalled but skipped in favour of clarity.

Indicators used

- **EMA 9 / 21 / 50** — multi-timeframe trend alignment (fast > mid > slow = uptrend).
- **RSI-14** — momentum confirmation; refuses overbought/oversold extremes (52–78 long, 22–48 short).
- **VWAP proxy** — price must be on the correct side of value.
- **ATR-14** — volatility, used to size iron-condor/fly wings and scale trend strength.
- **Opening Range (09:15–10:00 IST)** — first-bar high/low for breakout detection.

The seven outcomes (priority order)

Strategy	Setup	Trigger condition	Option structure
ORB_CE	Opening-range breakout (bull)	Spot breaks above OR-High in the morning window with bullish EMA/VWAP stack	Buy ATM CE
ORB_PE	Opening-range breakdown (bear)	Spot breaks below OR-Low in the morning window with bearish EMA/VWAP stack	Buy ATM PE
IRON_FLY	Expiry-day theta harvest	Expiry day + trend strength < 28 (price pinned, premium decaying)	Sell ATM CE+PE, buy ±300 wings
DIRECTIONAL_CE	Strong uptrend	Trend strength ≥ 50 and full bullish alignment	Buy ATM CE
DIRECTIONAL_PE	Strong downtrend	Trend strength ≥ 50 and full bearish alignment	Buy ATM PE
IRON_CONDOR	Range-bound / choppy	Trend strength < 30, EMAs flat	Sell ±200 strangle, buy ±400 wings
NO_TRADE	Mixed / unclear	None of the above — stand aside	—

Regimes: TRENDING_UP · TRENDING_DOWN · RANGE_BOUND · EXPIRY_LOW_VOL. A PCR (put-call ratio) alignment filter and a real-LTP check gate every live entry on top of the signal.

2 · Scalping Challenge (1-Minute System)

*The engine behind the 90-day SCALP challenge (**options.ScalpAnalyze**). A pure trend-following scalp on 1-minute NIFTY candles using just two moving averages plus a stochastic trigger. It outputs an ATM CE (long) or ATM PE (short) so it slots into the same challenge entry path.*

Step	Rule
Chart	1-minute NIFTY candles (needs ≥ 205 bars to seed EMA200)
Trend filter	Price above BOTH EMA50 & EMA200 → longs only; below both → shorts only; in between → no trade
Trigger	Stochastic %K(14,3): BUY when %K crosses up through 20 (in uptrend); SELL when %K crosses down through 80 (in downtrend)
Entry	Fires immediately on the confirming candle
Stop / target	Swing-based stop with a 1.5 : 1 reward-to-risk target, applied by the challenge/backtest caller

Why two EMAs + stochastic: the long-period EMAs keep the scalper on the right side of the dominant move, while the stochastic times pullback entries — avoiding chop between the averages.

3 · Daily Strategy Classifier

options.AnalyzeDaily — *the same regime logic adapted to DAILY candles for the 1-year backtest where intraday data isn't available. Uses EMA20 / EMA50 trend, ATR-14 regime detection and a 5-bar weekly-momentum check.*

- **Bullish:** $\text{EMA20} > \text{EMA50}$, $\text{spot} > \text{EMA20}$, and positive weekly change.
- **Bearish:** $\text{EMA20} < \text{EMA50}$, $\text{spot} < \text{EMA20}$, and negative weekly change.
- **Trend strength:** $\text{EMA separation} \div \text{ATR}$, boosted/penalised by EMA20 slope and weekly momentum.
- Feeds the same DIRECTIONAL / IRON-CONDOR / NO-TRADE structures as the intraday engine.

4 · Book-Based Strategy Suite

Seven strategies, each lifted from a published trading book and adapted to NIFTY options on a shared signal interface so they can be run in a like-for-like comparison backtest on identical data ([book_strategies.go](#), [booming_bulls.go](#)).

#	Strategy	Source / author	Core logic
1	ORB	Toby Crabel	First 15-min bar sets the opening range; a breakout bar that closes outside it triggers a directional entry (must close beyond, not just touch).
2	EMA Crossover	Stan Weinstein	Stage-analysis trend following via fast/slow EMA crossovers to ride sustained directional moves.
3	VWAP Reversion	Brian Shannon	Fades stretched moves back toward VWAP as a mean-reversion anchor for intraday price.
4	Short Straddle (Theta)	Lawrence McMillan	Sells ATM premium to harvest time decay when volatility/trend is low; defined by a neutral structure.
5	ATR Momentum Breakout	Van K. Tharp	Volatility-expansion breakout: enters when price thrusts beyond an ATR-scaled band with momentum.
6	SuperTrend	Olivier Seban	ATR-based SuperTrend flips define trend direction and entry side.
7	Booming Bulls	Anish Singh Thakur	Pure price-action: 15m→1H top-down trend, unmitigated supply/demand zones, candlestick confirmation, trade only with the 1H trend.

Each returns a uniform signal (direction, option side, 0–100 confidence, one-line reason), enabling a fair head-to-head backtest.

5 · Triple-Axiom Configuration

A configurable SuperTrend + EMA-pullback trade system ([triple_axiom_strategy.json](#)). Three 'axioms' combine: a SuperTrend regime, an EMA9/20 trend filter, and a timed pullback entry with fixed risk and a 2R target.

Component	Setting
SuperTrend	ATR period 10, multiplier 3 — defines the prevailing trend
EMA filter	Fast EMA 9 over slow EMA 20 — direction must agree with SuperTrend
Pullback entry	Wait up to 5 bars for a pullback to the EMA (0.15% touch tolerance); optional candle-colour confirmation
Risk / target	1% risk per trade, 2R target, swing-lookback 10 for stop placement
Gating	Minimum confidence 60; only trades 09:30–15:00 IST; needs ≥ 30 candles

6 · Equity & SIP Screeners (Alpha Engine)

Beyond index options, the alpha package screens individual stocks (India & US) across four horizons. Each screener emits signals with entry, target, stop-loss, risk-reward and a confidence score.

6.1 · Intraday signals

Strategy	Logic
VWAP_Reversion	Long mean-reversion back toward VWAP in an up-context
VWAP_Short	Short fade toward VWAP in a down-context
ORB_Long / ORB_Short	Opening-range breakout / breakdown entries
Momentum_Breakout	Trend-thrust breakout with momentum confirmation
Volume_Spike_Long / Short	Entries on abnormal volume surges in trend direction

6.2 · BTST (Buy-Today-Sell-Tomorrow)

End-of-day signals (after 15:00 IST) on stocks likely to gap up or continue next morning. Entry near close, exit next day 10:00–15:00 IST, stop below today's low. Three variants:

Variant	Trigger
BTST Momentum	Strong closing momentum into the bell
BTST Volume Surge	Breakout backed by an above-average volume spike
BTST Trend Continuation	Trend-aligned continuation with a controlled SMA20 gap

6.3 · Investment signals (swing & long-term)

Strategy	Horizon	Logic
Swing_Pullback	swing (1–4 wks)	Pullback into support within an uptrend
Trend_Following	swing	Aligned continuation of an established trend
Golden_Cross	longterm	50/200 moving-average golden cross
Value_Growth_Hybrid	longterm (3–12 mo)	Blends valuation and growth/quality scores

6.4 · Undervalued screener & 6.5 · Long-term US SIP

- **Undervalued:** composite value score (0–100) from fundamentals (P/E, P/B, EPS growth, ROE, debt/equity, dividend yield) plus a technical overlay, estimating fair value and % upside; tags each as NEW_BUY / ADD_MORE / WATCH.
- **Long-term US SIP:** scores US stocks for a 3-year monthly SIP via an overall SIP score (0–100) built from fundamental (0–35), valuation (0–25) and technical (0–10) components, with a growth-sector tilt and a 3-year CAGR target band.

Note: strategies are signal generators / analysis only — they recommend structures and scores; order execution, risk caps and the live/paper trailing-stop & daily-target controls are handled separately by the challenge service. This document is a code reference, not trading advice.